

FIELD BUILD MANUAL / REV 1.0 / 18 AUG 2026

ARC COMPANION TACTICAL DEVICE

ARDUINO UNO R3 BUILD GUIDE

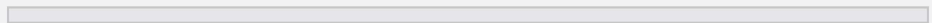
BUILD TIME 1 WEEK

COST 130 GBP

PLATFORM ARDUINO UNO R3

A fully Arduino-controlled tactical gaming device with military aesthetic, LED underglow lighting, responsive touchscreen interface, blueprint tracking for ARC Raiders, raid timer with audio/visual feedback, and portable battery-powered operation.

0% 0 / 111 STEPS



BOM 0%

SHOP 0%

PRINT 0%

WIRING 0%

ASSEMBLY 0%

CODE 0%

FINISH 0%

TEST 0%

RESET

BILL OF MATERIALS

0% ORDERED

ORD	ITEM	QTY	COST GBP	SOURCE
☐	7-inch HDMI LCD Touchscreen	1	55	Amazon UK
☐	5V Power Bank 10000mAh	1	20	Amazon UK
☐	WS2812B LED Strip 5m	1	15	Amazon
☐	USB Hub (Powered)	1	8	Amazon UK
☐	Black PLA Filament 500g	1	10	Amazon UK
☐	Spray Paint Kit	1	15	Local
☐	Hardware and Fasteners	1	5	Local

ORDERED SO FAR

0 GBP

TOTAL BOM

128 GBP

FROM YOUR PARTS STASH –
ALREADY OWNED

Arduino Uno R3 (x3) · Jumper Wires ·
Resistors · Tactile Buttons ·
Breadboards · USB Cables · LEDs ·
Capacitors · QIDI Q2 3D Printer. Total
value: 200+ GBP

PROJECT PLANNING NOTES

SHOPPING CHECKLIST

0% RECEIVED



HIGH · 55 GBP

7-inch HDMI LCD Touchscreen

Amazon Prime



HIGH · 20 GBP

5V Power Bank 10000mAh

Amazon Prime



HIGH · 15 GBP

WS2812B LED Strip

Amazon / Aliexpress



HIGH · 8 GBP

USB Hub (Powered)

Amazon Prime



MEDIUM · 10 GBP

Black PLA Filament

Amazon Prime



MEDIUM · 15 GBP

Spray Paint Kit

Local Hardware



LOW · 5 GBP

Mounting Hardware

Local Hardware

Budget summary: Essential 98 GBP, Optional 30 GBP, You Have 200+ GBP. Total Project Cost: 128-130 GBP (versus 220 GBP for Raspberry Pi).

3D PRINTING SETUP

0% DONE

STEP 1 · PREPARE DESIGN FILE

- ☐ Download OpenSCAD from openscad.org
- ☐ Download case file: case_design_arduino.scad
- ☐ Open in OpenSCAD
- ☐ Press F5 to preview
- ☐ Export to STL as arc_case.stl

STEP 3 · POST-PRINT PROCESSING · 1 H

- ☐ Cool completely (30 min)
- ☐ Remove from bed
- ☐ Remove supports with pliers
- ☐ Sand with 200-grit sandpaper (5 min)
- ☐ Smooth edges with 400-grit (10 min)
- ☐ Wipe clean with damp cloth
- ☐ Let dry completely (15 min)

STEP 2 · QIDI Q2 PRINT SETTINGS

- | | | |
|--------------------|-----------------|--------------------------|
| Nozzle Temperature | 200 °C | ☐ |
| Bed Temperature | 60 °C (PLA) | ☐ |
| Filament Type | Matte Black PLA | ☐ |
| Layer Height | 0.2 mm | ☐ |
| Infill Density | 15 % | ☐ |
| Support Material | Yes | ☐ |
| Print Speed | 50-60 mm/s | ☐ |
| Estimated Time | 12-14 hours | ☐ |

3D PRINTING NOTES – print issues, adjustments

HARDWARE WIRING

POWER WIRING

- ☐ Power Bank (5V/2A) connects to USB Hub
- ☐ Hub Port 1 to Arduino (Micro-B)
- ☐ Hub Port 2 to 7-inch LCD
- ☐ LED Strip to 5V direct

Total draw: 700–1400 mA. Use 2A+ power supply minimum.

BUTTONS · NO RESISTORS NEEDED

- ☐ Pin 2: MENU button to GND
- ☐ Pin 3: SELECT button to GND
- ☐ Pin 4: BACK button to GND
- ☐ Pin 5: ACTION button to GND

All buttons use internal Arduino pull-ups. Testing: press each button – Serial Monitor should show activity.

STATUS LEDs

- ☐ Pin 6 · 220 Ω · Red LED to GND
- ☐ Pin 7 · 220 Ω · Green LED to GND
- ☐ Pin 8 · 220 Ω · Blue LED to GND
- ☐ Pin 9 · 220 Ω · Yellow LED to GND
- ☐ Pin 11 to Piezo Buzzer (optional) to GND

Resistors protect Arduino pins from LED current.

WIRING CHECKLIST AND NOTES

HARDWARE ASSEMBLY

OPEN LAYOUT GUIDE

0%

01

MOUNT ARDUINO 15 MIN

- ☐ Place on breadboard for stability
- ☐ Use double-sided foam tape to secure
- ☐ Ensure all pins accessible
- ☐ Check no cables blocking ports

02

WIRE BUTTONS 20 MIN

- ☐ Connect all 4 buttons to GND line
- ☐ Connect button pins 2, 3, 4, 5
- ☐ Test each button
- ☐ Use breadboard for clean connections

03

INSTALL LEDs 15 MIN

- ☐ Solder 220 Ω resistors to each LED cathode
- ☐ Connect anodes to Arduino pins 6, 7, 8, 9
- ☐ Connect all cathodes to common GND
- ☐ Test that LEDs light

04

POWER DISTRIBUTION 15 MIN

- ☐ Connect USB hub to power bank
- ☐ Connect Arduino USB to hub
- ☐ Connect LCD USB to hub
- ☐ Connect LED strip 5V to power bank
- ☐ Test voltage with multimeter

05

CABLE MANAGEMENT 10 MIN

- ☐ Use zip ties to bundle cables
- ☐ Keep power and signal cables separate
- ☐ Route cables through case channels
- ☐ Label cables for identification

ASSEMBLY PROGRESS AND NOTES

ARDUINO CODE & TESTING

0%

STEP 1 · INSTALL ARDUINO IDE · 5 MIN

- ☐ Download from arduino.cc
- ☐ Install on your computer
- ☐ Launch Arduino IDE
- ☐ Tools > Board > Arduino Uno
- ☐ Tools > Port > USB COM port

STEP 2 · UPLOAD CODE · 10 MIN

- ☐ Open ARC_Companion_Arduino.ino
- ☐ Copy code into IDE
- ☐ Click Verify button
- ☐ Plug Arduino into computer
- ☐ Click Upload – expect “Done uploading”

STEP 3 · SERIAL MONITOR TEST · 5 MIN

- ☐ Tools > Serial Monitor
- ☐ Set baud rate to 9600
- ☐ Should see ARC COMPANION message
- ☐ Press MENU – Serial shows “Menu”
- ☐ Press SELECT – shows “Select”
- ☐ LEDs flash on startup

CODE AND TESTING NOTES

CASE FINISHING

0%

CASE PAINTING

- ☐ Mask screen opening and all ports with painter tape
- ☐ Light sand with 400-grit
- ☐ Apply 2 thin coats MATTE BLACK (5 min between)
- ☐ Apply METALLIC GRAY accents
- ☐ Apply 1 coat CLEAR POLYURETHANE
- ☐ **DRY 24 HOURS before assembly**

FINAL ASSEMBLY · 1 H

- | | |
|---|--|
| ☐ Verify all components working | ☐ Power on: all LEDs light |
| ☐ Press buttons: Serial responds | ☐ Test timer |
| ☐ Mount Arduino to case bottom | ☐ Install LED underglow strip |
| ☐ Mount LCD screen in bezel | ☐ Route all cables |
| ☐ Install acrylic diffuser | ☐ Close case carefully |
| ☐ Install 4 rubber feet | ☐ Final power test |

FINISHING NOTES

SIGN-OFF

FINAL TESTING

0% PASS

PASS	TEST	EXPECTED RESULT
☐	POWER ON	Arduino blue LED lights
☐	USB HUB	Hub LED lights
☐	7-INCH LCD	Screen boots
☐	ALL BUTTONS	Serial responses correct
☐	ALL LEDS	All illuminate correctly
☐	LED STRIP	Underglow lights
☐	BLUEPRINT SEARCH	Finds items correctly
☐	TIMER	Counts seconds
☐	OVERALL DEVICE	All systems operational

BUILD COMPLETION NOTES

BUILD COMPLETION LOG

0% COMPLETE — 0 OF 111 STEPS

Project complete at 100%: you have successfully built a custom ARC Companion tactical device.